

IoT for AAL: an Architecture via Information-Centric Networking

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Abstract—Internet of Things (IoT) systems have become a main part in future internet research. In IoT, heterogeneous devices are connected to sense the environment or to observe special tasks. Many research fields use the seamless IoT infrastructure to interact between the integrated devices and diverse services. Considering the healthcare research, the number of elderly people is rapidly growing and they need more support in their life. Therefore, a new technology called Ambient Assisted Living (AAL) has been proposed to better increase the life quality of elderly people. In AAL, the devices are mostly connected wirelessly to each other and/or to a healthcare service centre. It is challenging to build such a system and its deployment should be seamless and flexible. Therefore, IoT provides all main characteristics necessary for AAL environment and applications, including heterogeneous devices. However, IoT devices are resource-constrained with energy and memory.

Recently, Information-Centric Networking (ICN) has been established as a best communication paradigm for future internet, particularly in IoT. Based on hierarchically structured names, ICN matches the application pattern of IoT systems and provides an efficient and intelligent communication paradigm for both IoT and AAL (IoT-AAL) technologies. In this work, we design an IoT architecture for AAL via ICN approach which will effectively improve the performance of communication between the heterogeneous devices and services integrated in IoT-AAL.

Index Terms—Internet of Things, Ambient Assisted Living, Information-Centric Networking, Named Data Networking.

I. INTRODUCTION

Internet of Things (IoT) is a promising technology to increase the comfort and quality of life and opens new ways of interaction between people and things. In IoT systems:

- i) devices communicate with each other through the Internet.
- ii) devices are equipped with sensors that sense the environment.
- iii) data collected from sensors are reachable from any place in the world.
- iv) devices can communicate and learn from each other.

Real life applications in healthcare sectors, home automation, industry, smart cities and monitoring scenarios benefit from the low-cost wireless technology in IoT on one hand [1]. On the other hand, IoT system has many challenging features: many devices are resource-constrained with energy and memory, are highly heterogeneous and their applications continuously transmit transient information. Since the heterogeneous devices in IoT are limited with energy and memory, requesting, delivering and updating the information is a big challenge in such networks.

The main goal of IoT is to connect things together and people so that people can deliver information anytime and anywhere. Furthermore, IoT has become important in future Internet research to easily create a system to help people in their life routine through small smart devices, especially in healthcare scenarios [2].

The idea to use IoT concept for Ambient Assisted Living (AAL) has already been discussed in [2] and [3]. AAL encompasses assistance systems and services which deal to support old people or people with disabilities in their life. AAL technology uses different smart devices and healthcare tools for building and developing the system, for instance in home environment [4] or nursing home. Such smart devices and tools are connected to each other and/or to the healthcare centre. Furthermore, these devices are equipped with sensors to observe the healthcare state of elderly and forward such information or results to the responsible healthcare point/person. Additionally, AAL is trying to increase the life quality of elderly people using the different smart and efficient devices.

Using IoT for AAL (IoT-AAL) should simplify the interaction between diverse healthcare devices and healthcare centre/person. Furthermore, IoT provides all main characteristics necessary for AAL environment including heterogeneous devices. We believe that IoT-AAL increases the safety and efficiency in AAL fields for elderly, e.g. in home environment, emergency systems or healthcare monitoring. Building and developing such

systems is challenging and is related to save people life.

Therefore, the research community is trying to define a good networking solution for IoT. Several researches are considering the Information-Centric Networking (ICN) as potential communication paradigm for the future Internet [5], which replaces the host-to-host communication paradigm with information-centric paradigm. Since ICN matches the pattern of information-oriented applications, it is considered as a good solution for the IoT [6]. Named Data Networking (NDN) is one of the latest and important ICN approaches which use named content to deliver the data. In section III, we introduce the NDN architecture and give an example.

To achieve effective IoT-AAL system in different applications scenario we propose an architecture via NDN. The main concepts like naming, routing and forwarding in NDN system are also considered, including the IoT infrastructure.

The rest of the paper is organised into four sections as follows: Section 2 describes the IoT and AAL System and gives a motivation example for the IoT-AAL; Section 3 discusses the ICN communication paradigm and the integration of NDN in IoT-AAL systems. The proposed architecture and its components for IoT-AAL system are presented in section 4. Finally, section 5 concludes the paper and outlooks the future work.

II. IoT-AAL SYSTEMS

A. IoT-AAL Systems

Recently, the research communities are focusing on research of IoT as the best infrastructure for integration of million devices on the Internet. Since IoT is designed to easily integrate and interact heterogeneous devices, it has become a main part in research for future Internet.

The goal of AAL is to promise many opportunities for elderly people using different healthcare services and devices (most small and wireless devices). Most of these devices are connected wirelessly to exchange information or state about patients or elderly people. However, sensing, updating and requesting current sensor information in healthcare services and IoT-AAL applications is challenging.

Regarding the research on AAL and its application, some works address it. [7] discusses and reviews diverse AAL applications in smart homes. The authors show the role of safety and security for future IoT in healthcare services. Other work [8] proposes an architecture based on a service-oriented architecture (SOA) to support AAL environment integrated in body sensor networks.

However, other works [2] and [3] show that IoT infrastructure matches better for AAL systems. [2] discusses also the integration of IoT and shows some benefits of its communication in AAL paradigm. The authors believe

that the combination of Keep in Touch (KIT) technology and Healthcare services for elderly people could be used to build AAL systems through IoT infrastructure.

In [3], the authors discuss the future of computing and communication based on IoT and its challenges. They present a platform of cloud-based IoT to manage the integration of devices via AAL applications. For that, they introduce an architecture to integrate heterogeneous devices via a gateway using a service-oriented standard environment, and framework based on cloud computing to collect data from the sensor through gateways.

IoT system provides many features for AAL in different application scenarios like telemonitoring and people-to-people communication (with relatives or care organisations). In this work, we propose a new architecture for IoT-AAL via ICN communication paradigm to simply solve all issues presented above. More about ICN can be found in section III. To the best of our knowledge our work is the first work which is presenting seamless and flexible architecture using a new and efficient communication paradigm based on information.

B. Example Motivation

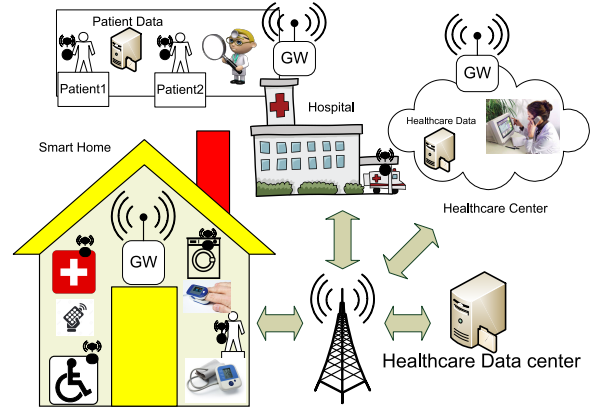


Fig. 1. IoT AAL scenario

In this section we want to give an example motivation in IoT for AAL scenario. Figure 1 demonstrates the integration with a set of smart heterogeneous devices in a smart home which are connected to monitoring systems. Some of the smart devices are sensing and measuring basic state of elderly people, such as temperature or heart rate, and are wirelessly connected to the Gateway. They are used as healthcare tools for elderly person which are sending information of patient state to the healthcare service centre based on the device configurations set by, e.g. a doctor. Furthermore, as shown in Figure 1 we consider the hospital and Healthcare Centre as healthcare service centre which have access to the healthcare centre

data. A responsible person in the healthcare centre (e.g. a doctor or nurse) can interact directly by contacting the healthcare devices located on the patient to send important information related to the elderly or patient person. Additionally, the healthcare centre can manage the monitored information which are stored from another organisation. The sensors and devices connected to a monitoring system in this scenario benefit from IoT infrastructure in term to detect and avoid any emergency situation.

III. IoT-AAL VIA ICN

First, this section gives motivation about the ICN approach and introduces the Named Data Networking (NDN) as ICN communication paradigm. Second, we will introduce our lightweight and flexible architecture for IoT-AAL system based on NDN.

A. Information-Centric Networking (ICN)

ICN is a future network architecture which has achieved a significant interest from the research community. ICN replaces the host-to-host communication paradigm of traditional TCP/IP protocol to information-centric communication. With ICN, the communication between devices is based on the information. More information about ICN architecture can be found in [5]. In the next section we refer to the NDN architecture which is one of the newest and important ICN approach.

Named Data Networking (NDN): NDN is a new networking paradigm for future Internet that uses hierarchically structured names to deliver content. The names are carried in so-called Interest and Data Packets. NDN nodes maintains three tables:

- i) Content Store (CS) to cache in-coming data.
- ii) Pending Interest Table (PIT) to keep track of the forwarding data.
- iii) Forwarding Information Base (FIB) as a routing table based on content names.

For requesting data from producer the consumer sends an Interest which carries the content/data name. If the producer receives the Interest, it first looks in its CS; if a match is found, then the node sends the data back. Otherwise, if there is a matching entry in the PIT, the Interest is discarded because an equal request has already been forwarded. Otherwise, a new PIT entry is created and the Interest is forwarded to the matched interface stored in FIB. The Data packet follows the chain of PIT entries back to the consumer and intermediate nodes cache a copy of the delivered content in its CS which reduces and limits the massive data access to the same producer. Further information about NDN can be found in the NDN project [9].

B. NDN for IoT-AAL

Recently, NDN has been proposed for future Internet in many research papers so far, but no work of them addresses it for the IoT-AAL and its limited features. NDN is actually designed for static, large and popular content in the traditional Internet. Usually, the content routers in NDN are fixed and wired connected and the delivery path of such contents is mostly static. The performance of such schemes in IoT causes mostly large overhead and complexity which make them unsuitable in presence of resource-constrained devices. Therefore, many works modify the standard implementation of NDN to match the resource-constrained IoT devices [10]. We also consider the mentioned limitations in IoT and design a high-level IoT-AAL architecture based on NDN, as shown in Figure 2.

The IoT applications are constantly producing data which are requested either from consumers in the same network or other network. This concept matches the NDN architecture. We believe that NDN matches the requirements of IoT system and uses its functionalities (data aggregation, routing, forwarding, security, etc.) to build robust and scalable IoT-AAL system.

As observed in [11], NDN can be very effective for IoT. Furthermore, the research results in [12] and [13] show the efficient and flexibility using NDN for heterogeneous devices in term of saving energy which are critical in IoT systems. Additionally, NDN can easily operate the link layer without requiring an overlay over the IP stack.

IV. THE PROPOSED ARCHITECTURE

Our Architecture in Figure 2 is proposed for IoT-AAL systems based on the hourglass architecture of NDN. In this architecture, NDN is extended to meet the requirement of IoT-AAL systems.

A summary of the main components in our architecture is defined in Table I. In the following we present the functionalities of every component in more details.

A. Management

The management system proposed in this architecture contains Naming, Routing and Forwarding mechanisms which will be discussed in the following:

1) **Naming:** The concept of requesting data in NDN is based on a hierarchically structured name. The names contain human friendly components and are location independent unlike IP approach. NDN's name structure allows applications to control the data transmission between producers and consumers more flexible. However, the same generated data from producer could be interesting for more than one consumers at the same time (e.g. care organisation and a doctor). Therefore,

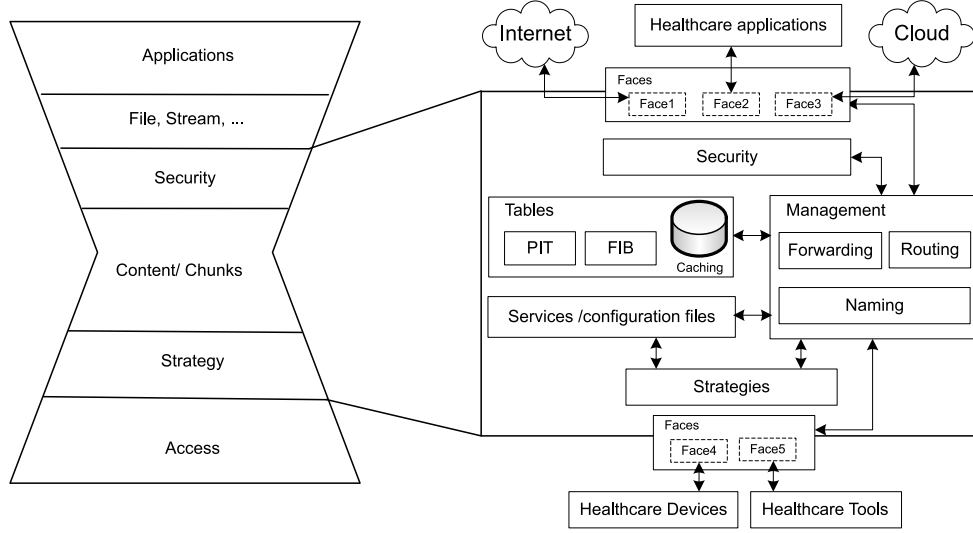


Fig. 2. NDN Hourglass Architectures (left) and proposed IoT-AAL Architecture via NDN Approach (right)

NDN enables that intermediate nodes can save a copy of transmitted data. Now, the consumers are able to retrieve the generated data from any nodes in the network which has a copy of them without the need to forward the interest to the producer. Such mechanisms are beneficial for IoT systems because of reducing the traffic load in the network.

Mostly in IoT systems, the names describe a special task, events or application scenarios. If we consider the naming concept in IoT-AAL, for instance, an Interest name to request information about heart rate sensor located on patient could look like in Figure 3, where ”/” indicates the components and does not belong to the names. The name here contains different components as shown in Figure 3 and described in table II.

Using the proposed name structure, the IoT-AAL applications can easily understand the request and its answer to facilitate data transmission [11]. NDN concept and its name structure match the principle of IoT communication. Since the IoT system requests data using a human-friendly names (clear names), this concept is beneficial for the IoT system but has some security gaps. Some of the security aspects will be discussed in section IV-D.

2) **Routing and Forwarding:** Routing and forwarding the packets in a NDN node is based on the names. In NDN, the consumers do not need to know the address of devices in order to offer and/or request data. Furthermore, unlike traditional IP, NDN mobile devices do not need to reconnect their communication because of changing the addresses, names remain the same [9].

Every NDN node has the name prefix of the data which can be covered and served on this node. The

node announces the network about the name prefixes via routing protocol and every node builds the FIB entry based on received information. After building the FIB entry for the received prefix the node floods it to the neighbour nodes. To keep the FIB table sizes scalable to the number of data names, NDN proposes efficient approaches for that, see NDN project description [9].

Naming scheme in NDN should solve different problem like data aggregation in IoT environment. The research in [14] shows how efficient and useful data aggregation in Wireless sensor networks via NDN. They define a unicast faces for packet filtering to reduce the number of transmission in the network. Moreover, they design a forwarding services to create network overlay structures by applications.

Many developed approaches propose routing and forwarding strategies for wireless network environment based on NDN name scheme. The authors in [15] show that using such named-based routing approaches can achieve a high performance on the network because of the intelligent and adaptive of forwarding strategies in NDN.

B. Caching

Caching in IoT-AAL devices is challenging because of the small cache size and their device limitation. Different caching strategies are used to provide availability and reliability in ICN-IoT systems. Many caching strategies (e.g, probabilistic caching and caching everything) are developed based on the application and data exchange pattern, Caching everything strategy is the standard caching strategy in NDN.

In-network Caching is one of the main concept of NDN, IoT could benefit from in-network caching in

TABLE I
CONTENT AND DESCRIPTION OF THE MAIN COMPONENTS IN OUR PROPOSED ARCHITECTURE

Components	Contains	Description
Management	Routing, Forwarding and Naming mechanisms	To manage the routing and forwarding by receiving NDN packets (Interest/Data) based on Names
Tables	CS, PIT and FIP tables	To save entries created by the Management based on received Interest and Data
Faces	Multiple Faces for different Applications	To send and receive incoming and outgoing data from/in the network, an application can contains many faces with different name prefixes
Security	Security Algorithms	To secure transmitted data and Interests
Strategies	e.g. Forwarding strategies	To take best decision about forwarding NDN packets
Services and Configuration	Setup parameters	To save some parameters used for configuration the services

order to reduce the number of hops replication, content delivery time and amount of traffic. Additionally, the data in NDN are cached in the network layer and the same data can be shared in the intermediate nodes, which cause to reduce the overhead traffic at the producer. The works [3] and [4] discuss the In-network caching issues in IoT systems and show its advantage in term to save energy, increase the availability and reduce the complexity. Considering the energy level on the node, a distributed probabilistic caching strategy is proposed to improve data retrieval performance and the energy efficiency.

Additionally, in-network caching has another advantage, the gateway can permanently save the transmitted data produced from devices in its range. This gateway can aggregate the received information and answer of received Interest without the need to forward it to the original producer.

C. Strategies

In this section, we introduce the forwarding Strategy in NDN which could be implemented in the Strategy layer. **Forwarding Strategy:**

Most of devices today are mobile and need multiple network interfaces (in NDN called faces) to avoid reconnection with producer by mobility. In NDN architecture the consumers benefit from the designed multiple faces, so by changing the connectivity on node, NDN can always exchange the data if is physically possible.

Packet forwarding in NDN is based on finding the best faces with the longest matched prefixes in the FIB table. The node chooses the best matched prefix entry from FIB table for forwarding packets through it. However, if the node lost the connectivity to this face, it can choose the second best face.

There are many forwarding strategies which have been developed to support such functionality like how to forward Interests. Some of them are *sendToAll*, *sendToBest*,

interestTimeDown that can be used for taking forwarding decision based on e.g, application.

Such actions and more are implemented in the strategies layer and are considered in our architecture for IoT-AAL devices. In fact, the default strategy in NDN is to send an Interest to all Broadcast faces, if no answer, try with all faces in sequence [9].

For IoT-AAL systems, efficient and robust forwarding strategies should be developed to avoid collision in broadcast medium. The work [15] has already discussed this issue and designed a forwarding strategy for wireless and mobile nodes in IoT environment.

D. Security

Considering the NDN security control for IoT, we define a component in our architecture for security and privacy mechanisms on devices attending in the networks.

Our Architecture provides NDN security mechanism to secure the transmitted data in IoT-AAL. Each transmitted NDN packet is signed with its name together. Applications in IoT-AAL benefit from the signed Interest in term to recognize the trusted and authorized consumer which is sending an Interest.

[16] considers applications in lighting controlling systems which send authenticated command with private key to so called fixture. The fixture checks the received command by verifying its signature, if the command is successfully verified the fixture sends an acknowledgement in form of signed content.

An efficient security mechanism for IoT-AAL system should be considered to reduce the computation and delay in constrained devices.

V. CONCLUSION

In this paper, we motivated the advantage of deployment Internet of Things (IoT) for Ambient Assisted Living (AAL). We studied how IoT can be beneficial

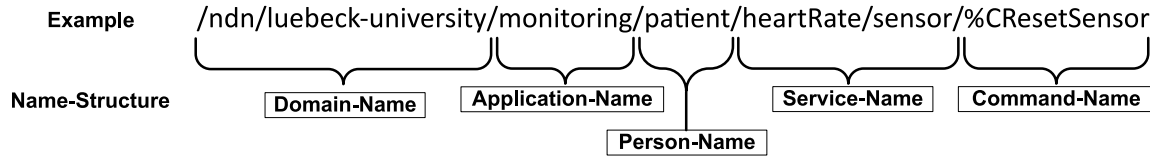


Fig. 3. Proposed Name Structure via NDN

TABLE II
DESCRIPTION OF NAME STRUCTURE IN OUR PROPOSED
ARCHITECTURE

Attributes	Description
Domain-Name	Domain name of organisation or centre which uses the IoT-AAL
Application-Name	To identify the application run by Domains
Person-Name	The name of patient or person who is interesting for, e.g. a doctor or has a healthcare device/Tool
Service-Name	The service which runs on the devices, configured by a doctor or healthcare center
Command-Name	Used to send specific information to control the device, e.g. send command to reset the device

for AAL (IoT-AAL) in terms of delivery performance and reducing the load on constrained devices.

Information-Centric Networking (ICN) approach is proposed as a new communication paradigm for IoT-AAL. Therefore, we consider the Named Data Networking (NDN) as one of the most important ICN instances. NDN communication paradigm should increase data availability and reduce the number of traffic in the network for heterogeneous devices used in IoT-AAL environments.

We propose a new architecture for IoT-AAL via NDN and show the robustness and efficiency by using our design. Furthermore, the proposed architecture contains different components to seamlessly manage routing, forwarding, naming and security mechanisms in IoT-AAL. The architecture can be easily and flexibly extended to add different components to improve the performance of IoT-AAL system if needed.

As a future work, we plan to implement and evaluate the proposed architecture in a real IoT-AAL system. Based on the evaluation result we will extend it to add new components and implement new routing and forwarding strategies for IoT-AAL.

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